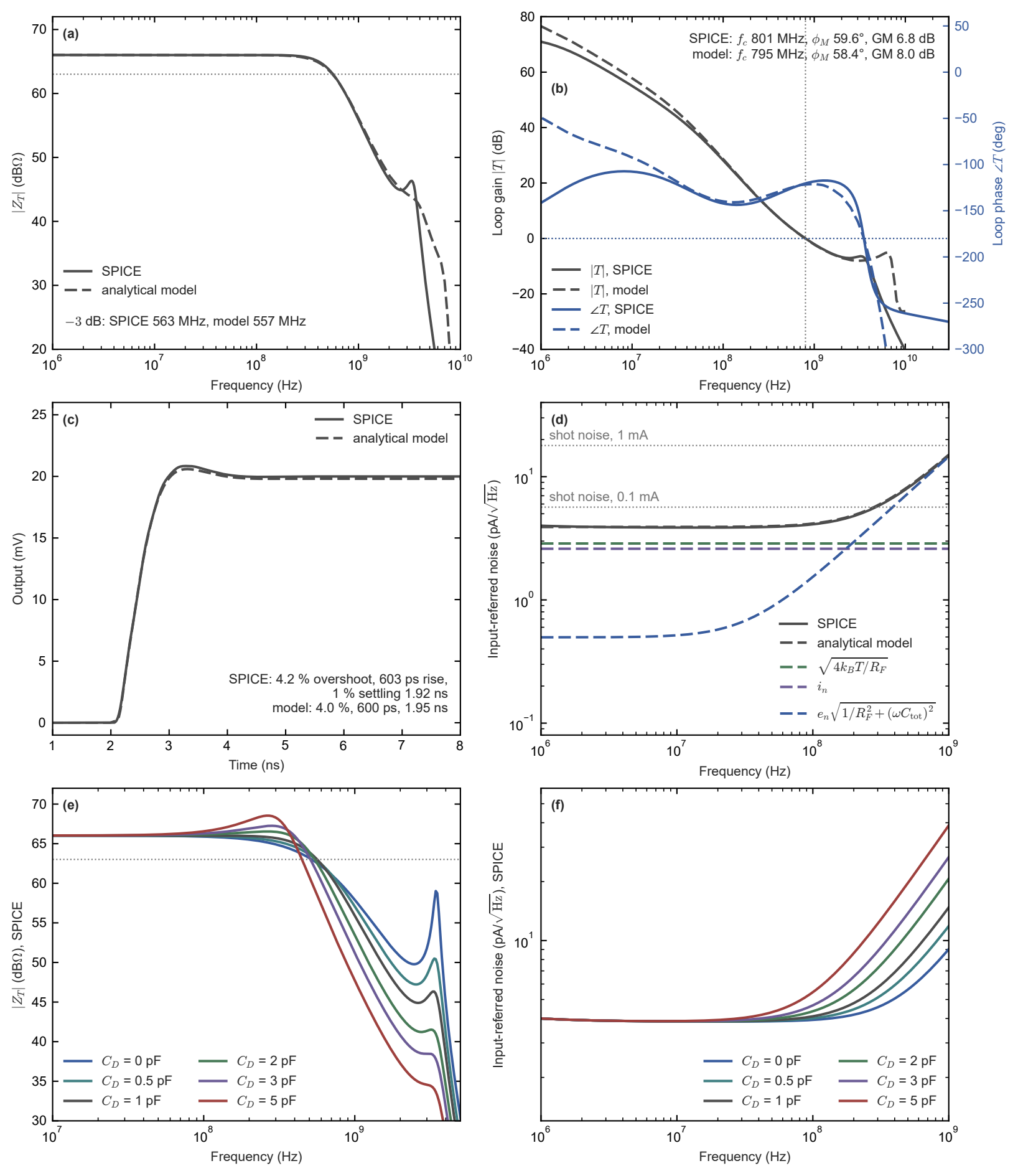

(a)
$|Z_T|$ (dBΩ)
Frequency (Hz)
SPICE
analytical model
−3 dB: SPICE 563 MHz, model 557 MHz
(b)
SPICE: f_c 801 MHz, ϕ_M 59.6°, GM 6.8 dB
model: f_c 795 MHz, ϕ_M 58.4°, GM 8.0 dB
Loop gain $|T|$ (dB)
Loop phase $\angle T$ (deg)
$|T|$, SPICE
$|T|$, model
$\angle T$, SPICE
$\angle T$, model
Frequency (Hz)
(c)
SPICE
analytical model
Output (mV)
SPICE: 4.2 % overshoot, 603 ps rise,
1 % settling 1.92 ns
model: 4.0 %, 600 ps, 1.95 ns
Time (ns)
(d)
shot noise, 1 mA
shot noise, 0.1 mA
Input-referred noise (pA/$\sqrt{\text{Hz}}$)
SPICE
analytical model
$\sqrt{4k_BT/R_F}$
i_n
$e_n\sqrt{1/R_F^2+(\omega C_{\text{tot}})^2}$
Frequency (Hz)
(e)
$|Z_T|$ (dBΩ), SPICE
C_D = 0 pF
C_D = 0.5 pF
C_D = 1 pF
C_D = 2 pF
C_D = 3 pF
C_D = 5 pF
Frequency (Hz)
(f)
Input-referred noise (pA/$\sqrt{\text{Hz}}$), SPICE
C_D = 0 pF
C_D = 0.5 pF
C_D = 1 pF
C_D = 2 pF
C_D = 3 pF
C_D = 5 pF
Frequency (Hz)